County-Level Socioeconomic Correlates of Violent Crime in the United States

A Data Analytics Approach Using SQL, BigQuery, Python, and Power BI

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Executive Summary

Violent crime remains one of the most significant public policy challenges in the United States. Public discussions frequently attribute crime to individual factors such as poverty, education, religion, or drug use, yet these explanations are often evaluated independently rather than within a comprehensive analytical framework.

The objective of this project was to investigate whether county-level socioeconomic characteristics are associated with violent crime rates across the United States. Publicly available datasets from the FBI, U.S. Census Bureau, CDC, County Health Rankings & Roadmaps, and the U.S. Religion Census were integrated into a unified analytical database using Google BigQuery.

More than 3,200 U.S. counties were analyzed using SQL, Python, exploratory data analysis, Power BI dashboards, Pearson correlation analysis, and multiple linear regression.

The analysis found that no individual socioeconomic variable served as a strong standalone predictor of violent crime. Food insecurity and poverty exhibited the strongest positive correlations with violent crime, while median household income demonstrated a modest negative relationship. Educational attainment, religiosity, smoking prevalence, population density, and drug overdose mortality exhibited little or no meaningful linear relationship individually.

A multiple regression model explained approximately 8 percent of the observed variation in violent crime rates, suggesting that violent crime is influenced by numerous interacting economic, demographic, institutional, and social factors beyond those examined in this study.

Introduction

Violent crime has profound social, economic, and public health consequences. It affects individuals, families, communities, businesses, and government agencies while influencing public perceptions of safety and quality of life. Understanding the

factors associated with violent crime remains an important objective for researchers, policymakers, and criminal justice professionals.

Numerous explanations for violent crime have been proposed throughout the academic literature and public discourse. These include economic inequality, educational opportunity, unemployment, population density, religiosity, substance abuse, and other social determinants of health. While many of these variables have been studied individually, comparatively few analyses integrate multiple national datasets into a unified county-level analytical framework.

This project sought to answer the following research question:

Which county-level socioeconomic characteristics are associated with violent crime rates across the United States?

Unlike many previous analyses that rely upon state-level statistics, this project intentionally selected the county as the unit of analysis.

State-level analyses provide approximately fifty observations but often combine urban, suburban, and rural populations exhibiting substantial socioeconomic variation. County-level analysis provides more localized socioeconomic context while increasing the sample from roughly fifty observations to more than 3,200 counties, substantially improving statistical power and geographic resolution. Although counties are not perfectly homogeneous, they provide a practical balance between data availability, analytical precision, and socioeconomic consistency.

Research Questions

This project investigated the following questions:

1. **Is educational attainment associated with violent crime?**
2. **Is household income associated with violent crime?**
3. **Is poverty associated with violent crime?**
4. **Does population density influence violent crime?**
5. **Is religiosity associated with violent crime?**
6. **Are food insecurity and other measures of socioeconomic distress associated with violent crime?**
7. **Are health-related variables such as smoking and drug overdose mortality associated with violent crime?**

Data Sources

Five primary public datasets were integrated into a unified county-level database.

FBI National Incident-Based Reporting System (NIBRS)

Provided incident-level crime records used to calculate county violent crime rates.

American Community Survey (ACS)

Provided demographic and socioeconomic variables including:

- **Population**
- **Educational attainment**
- **Household income**
- **Poverty**
- **Housing**
- **Population density**

County Health Rankings & Roadmaps

Provided county-level measures including:

- **Food insecurity**
- **Smoking prevalence**
- **Drug overdose mortality**
- **Health outcomes**
- **Health behaviors**
- **Social determinants of health**

CDC PLACES

Provided modeled county estimates including:

- **Depression**
- **Mental distress**
- **Obesity**
- **Physical inactivity**
- **Sleep**

- Additional behavioral health measures

U.S. Religion Census

Provided county estimates of religious adherence across the United States.

Data Preparation

Data engineering represented one of the largest components of the project.

The project required importing, cleaning, transforming, and integrating multiple national datasets within Google BigQuery.

Major preparation steps included:

- Importing over 53 million NIBRS incident records
- Importing ACS socioeconomic data
- Importing County Health Rankings
- Importing CDC PLACES data
- Importing the U.S. Religion Census
- Standardizing county FIPS identifiers
- Joining datasets into a unified county-level analytical table
- Engineering variables including violent crime rates per 100,000 residents
- Capping religiosity values that exceeded logical limits
- Filtering counties with populations below 10,000
- Excluding jurisdictions with zero reported violent crime to reduce the influence of likely incomplete NIBRS reporting

Exploratory Data Analysis

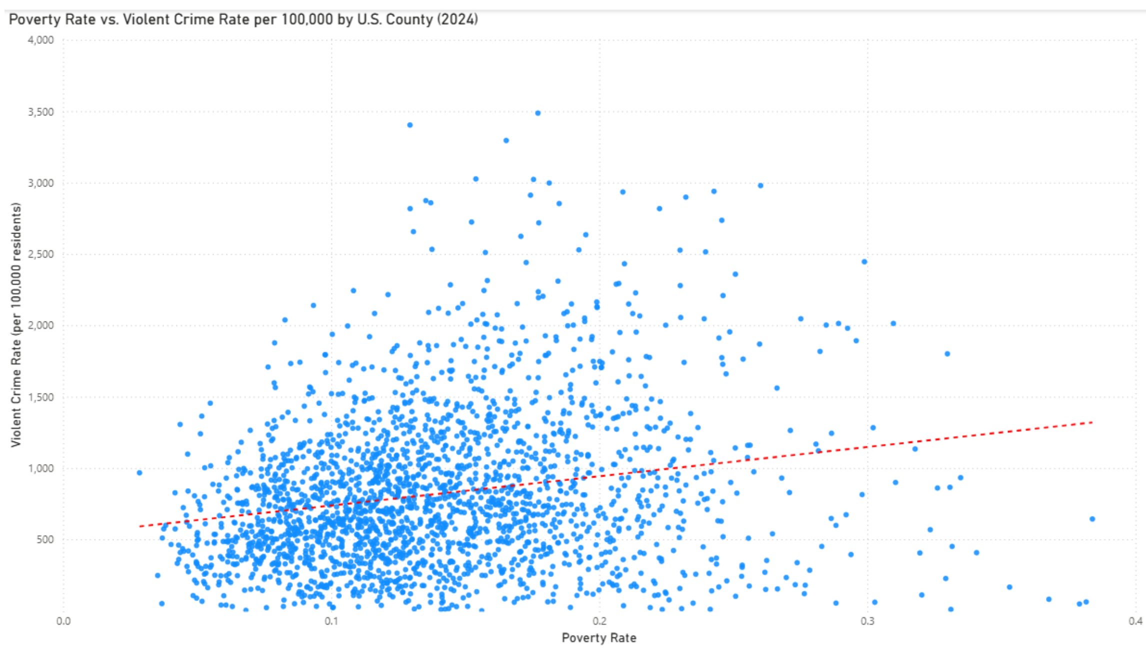
Exploratory Data Analysis (EDA) consisted of SQL queries, Python notebooks, and Power BI visualizations examining relationships between violent crime and selected explanatory variables.

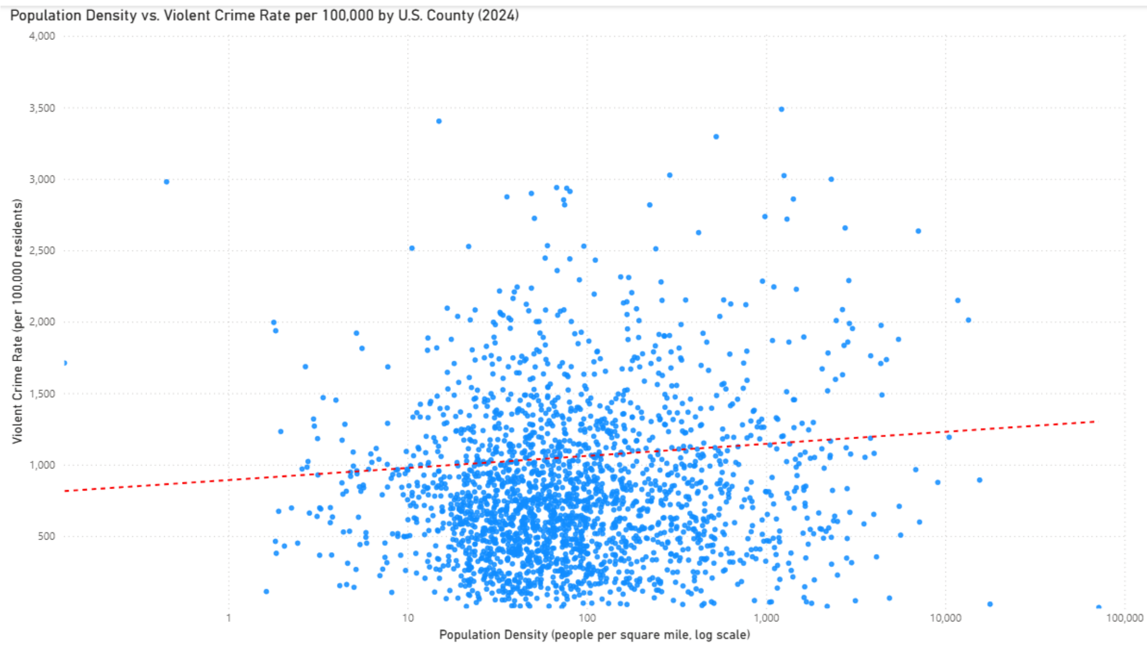
Scatterplots were created for:

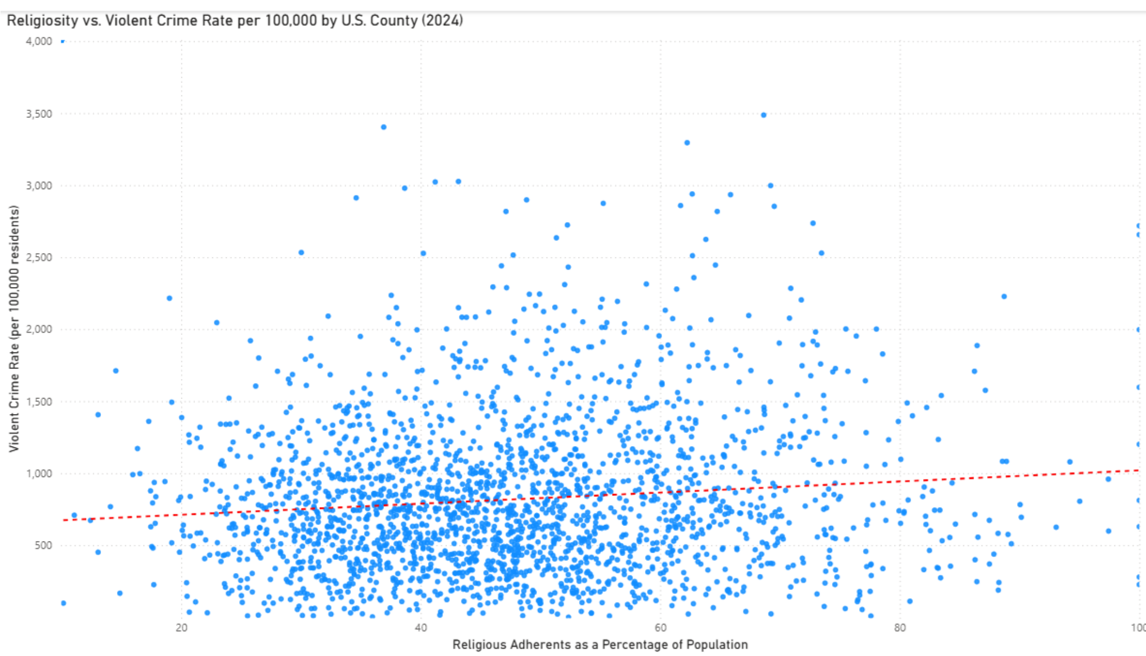
- Bachelor's degree attainment
- Median household income

- **Poverty rate**
- **Population density**
- **Religious adherence**
- **Food insecurity**
- **Smoking prevalence**
- **Drug overdose mortality**

Trend lines were added to each visualization to facilitate preliminary assessment of potential linear relationships.







Statistical Analysis

Pearson Correlation Analysis

Variable	Pearson r
Food insecurity	0.212

Variable	Pearson r
Poverty rate	0.208
Median household income	-0.148
Religious adherence	0.109
Population density (log)	0.096
Smoking prevalence	0.045
Bachelor's degree attainment	-0.019
Drug overdose mortality	-0.005

Food insecurity and poverty exhibited the strongest positive relationships with violent crime, although both remained statistically weak. Median household income demonstrated a modest inverse relationship with violent crime. Other variables exhibited only weak or negligible linear associations.

Multiple Linear Regression

A multiple linear regression model was developed using BigQuery ML.

Dependent Variable:

- Violent Crime Rate per 100,000

Independent Variables:

- Education
- Income
- Poverty
- Population density
- Religiosity
- Food insecurity
- Smoking
- Drug overdose mortality

Model Performance:

- **R^2 = 0.082**
- **Mean Absolute Error = 402.1**

The regression explained approximately 8% of county-level variation in violent crime, indicating that while the included variables contributed modest predictive value, most variation remains unexplained by this model.

Findings

Several important findings emerged.

- **No individual socioeconomic variable demonstrated a strong standalone relationship with violent crime.**
- **Measures of economic hardship—including food insecurity, poverty, and lower household income—exhibited the strongest associations.**
- **Educational attainment demonstrated little measurable relationship with violent crime.**
- **Religiosity exhibited only a weak positive association.**
- **Drug overdose mortality demonstrated essentially no linear relationship with violent crime.**
- **Population density exhibited only a weak relationship after logarithmic transformation.**

These findings suggest that violent crime cannot be adequately explained by any single socioeconomic variable.

Limitations

Several limitations should be acknowledged.

- **The analysis is observational and cannot establish causation.**
- **County-level averages may obscure important within-county variation (ecological fallacy).**
- **NIBRS reporting remains incomplete in some jurisdictions.**
- **Several explanatory variables are themselves correlated, complicating interpretation of individual regression coefficients.**
- **Numerous potentially important explanatory variables were beyond the scope of the present study.**

Future Research

The relatively low explanatory power of the regression model suggests that future research should incorporate a broader range of institutional, educational, familial, and community variables.

Potential educational variables include:

- **Per-pupil funding**
- **Student-to-teacher ratios**
- **Class size**
- **Graduation rates**
- **School climate**
- **Disciplinary policies**
- **School choice**
- **Educational opportunity**

Potential policing and criminal justice variables include:

- **Officers per capita**
- **Departmental funding**
- **Expenditures per officer**
- **Officer workload**
- **Fatigue and burnout**
- **Response times**
- **Clearance rates**
- **Community policing strategies**
- **Patrol deployment practices**
- **Use-of-force incidents**
- **Accountability measures**
- **Departmental militarization**

Future analyses should also investigate family and social support characteristics, including:

- **Family stability**
- **Parental involvement**
- **Household composition**
- **Residential stability**
- **Homeownership**
- **Multigenerational households**
- **Extended family support**
- **Mentoring relationships**
- **Community social capital**
- **Intimate partner violence**
- **Aggregate measures of adverse childhood experiences (where available)**
- **Availability of family support services**

Finally, expanding this project into a multi-year longitudinal analysis would allow researchers to examine how changes in socioeconomic conditions, institutional practices, family characteristics, and community environments relate to changes in violent crime over time.

Conclusions

This project demonstrated the feasibility of integrating multiple large-scale public datasets into a unified county-level analytical framework using Google BigQuery, SQL, Python, and Power BI.

The findings suggest that violent crime is a multifactorial phenomenon that cannot be adequately explained by any single county-level socioeconomic characteristic. Although food insecurity, poverty, and household income exhibited the strongest associations observed in this study, these relationships remained modest. Educational attainment, religiosity, smoking prevalence, population density, and drug overdose mortality demonstrated comparatively weak or negligible individual relationships.

The multiple regression model explained approximately 8% of county-level variation in violent crime, indicating that much of the observed variation is likely associated

with additional economic, demographic, institutional, educational, familial, and community factors not included in the present analysis.

Beyond its substantive findings, this project demonstrates an end-to-end analytics workflow involving large-scale data acquisition, data engineering, SQL, Python, statistical analysis, business intelligence dashboards, and professional communication of results. It provides a foundation for future research incorporating additional variables, longitudinal analyses, and more advanced statistical and machine learning techniques to improve understanding of the complex factors associated with violent crime.

Artificial Intelligence Assistance

Artificial intelligence (OpenAI ChatGPT) was used for brainstorming, SQL troubleshooting, editing, and report organization. All data preparation, analysis, interpretation, and conclusions were completed and verified by the author.

References (APA 7)

Federal Bureau of Investigation. National Incident-Based Reporting System (NIBRS).
U.S. Census Bureau. American Community Survey 2020–2024 5-Year Estimates.
County Health Rankings & Roadmaps. 2024 County Health Rankings National Data.
Centers for Disease Control and Prevention. PLACES: Local Data for Better Health.
Association of Religion Data Archives. U.S. Religion Census.
OpenAI. (2026). ChatGPT (GPT-5.5) [Large language model]. https://chatgpt.com/